

Basic Concepts of Digital Image Processing

Gurumeet Kumar
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1 Introduction

Digital Image Processing (DIP) is the field of computer-based computing to analyze, modify, and extract information from images. It plays an important role in healthcare, autonomous vehicles, surveillance systems, and modern artificial intelligence applications.

A digital image is represented as a collection of pixels arranged in rows and columns. Each pixel stores information about brightness or color. It can be formally defined as a two-dimensional light-intensity function $f(x,y)$, where the value at the coordinates (x,y) gives the brightness at that point.

2 Types of Digital Images

2.1 Binary Images

Each pixel of this image is black(0) or white(1). These images are commonly used in text recognition and simple shape analysis.

2.2 Grayscale Images

There is no color in this image. Each pixel has a value from 0 (pure black) to 255 (pure white). A standard grayscale image uses 8 bits per pixel, giving 256 possible shades.

2.3 RGB Images or Color Images

Color images are generally represented using three channels: Red, green, and Blue. Combining these channels allows for the representation of millions of colors.

3 Digital Image Processing Pipeline

A typical image processing system follows the steps below:

1. Image Acquisition
2. Image Enhancement
3. Image Restoration
4. Image Segmentation
5. Feature Extraction
6. Object Recognition

3.1 Image Acquisition

The image is captured using cameras, scanners, or medical imaging devices.

3.2 Image Enhancement

This improves visual quality such as increasing contrast, sharpening edges, noise reduction, etc. This is a subjective process. It makes an image more visually appealing to Human.

3.3 Image Restoration

Restoration recovers the original image from a degraded image using mathematical models, e.g. Motion blur removal.

There is a famous example: the early images of the Hubble Space Telescope were restored in this way after a flaw in its mirror was discovered.

3.4 Image Segmentation

Segmentation divides an image into meaningful regions.

For example, in a medical scan, segmentation can separate a tumor from surrounding tissue.

3.5 Feature Extraction

It simplifies the raw data. Instead of looking at millions of raw pixels, the computer extracts basic descriptors of the isolated object, like its size, shape, color, or edge boundaries.

3.6 Object Recognition

It labels the object. The computer takes the extracted features and assigns a final name to the object.

4 Important Image Processing Techniques

4.1 Gaussian Smoothing

Gaussian filtering is used to reduce noise before further processing.

Benefits:

- Reduces unwanted noise
- Improves edge detection results

4.2 Fourier Transform

The Fourier Transform converts image information from the spatial domain to the frequency domain.

Applications:

- Image compression
- Filtering
- Image reconstruction

4.3 Morphological Operations

Morphological operations analyze shapes within images.

Two important operations are:

- Dilation
- Erosion

These are useful for removing noise and improving object boundaries.

“`latex`

5 Computer Vision

Computer Vision is the field of teaching machines to interpret and understand images in a way similar to humans.

The field can be broadly divided into three levels:

5.1 Low-Level Vision

Low-level vision works directly with pixel values without attempting to understand the meaning of the image.

Examples include:

- Image resizing
- Grayscale conversion
- Edge detection
- Color segmentation
- Brightness and contrast adjustment

5.2 Mid-Level Vision

Mid-level vision connects images to the physical world.

Examples include:

- Panorama stitching
- 3D reconstruction
- Motion detection
- Optical flow

5.3 High-Level Vision

High-level vision focuses on understanding the meaning of an image.

Examples include:

- Image classification
- Object detection
- Face recognition

- Medical image interpretation

Computer Vision is widely used in healthcare, robotics, autonomous vehicles, and surveillance systems.

6 Human Vision

CV is about teaching machines to see like us. Studying how our eyes and brain work helps us design smarter programs.

6.1 Evolution of Eyes

The earliest visual systems could only detect the presence or absence of light. Over time, more advanced eyes evolved that could form detailed images of the surrounding world.

Human eyes belong to the category of complex refractive eyes, which use a lens to focus light onto the retina.

6.2 Rods and Cones

The retina contains two major types of photoreceptor cells.

- Rods
 - Responsible for low light and night vision
 - Highly sensitive to light
 - Cannot detect color
 - Saturate quickly in bright light, which is why bright light destroys night vision temporarily
- Cones
 - Responsible for color vision
 - Concentrated in the fovea eye's sharp vision center (1.5mm)
 - Allow humans to distinguish fine details

Humans possess three types of cones (Short, medium and long) that are sensitive to red, green, and blue wavelengths.

6.3 Brain Processing of Vision

Vision is not performed entirely by the eyes. The visual cortex (vc) takes up 30% of the cerebral cortex, and in sighted people, two-thirds of all electrical brain activity is devoted to vision in some way.

The brain processes the signals received from the retina and interprets them as meaningful objects, shapes, and colors.

Different regions of the brain are responsible for:

- Edge detection (Handled by V1 - primary vc)
- Shape recognition (Handled by V2 - secondary vc)
- Motion perception (Handled by Dorsal stream - the “where and how” system)

- Object identification (Handled by Ventral stream -the "what" system)

*** **BlindSight - A remarkable phenomenon**

People with severe damage to their visual cortex report seeing nothing. Yet when asked to navigate a hallway with obstacles, they can do so without consciously knowing how. Their dorsal stream still processes visual information and guides movement without it ever reaching consciousness.

6.4 Color and Light

Objects do not possess color by themselves. The perceived color depends on the light reflected from the object and the response of the cone cells in the eye.

Humans see wavelengths roughly from 400nm (violet) to 700nm (red) - the visible spectrum.

Our color perception is relative activation of those three types of cones. Each cone cell can sense about 100 different shades, giving roughly 1 million perceivable colours.

- **Metamers:** Sometimes different wavelengths produce identical cone responses and thus they look the same to us. This is why screens can fake any colour using just three LEDs.

6.5 Color spaces

- **RGB - The Cube:** It has 3 axes - red, green, blue. It is great for storing and displaying images.
- **HSV - The Cylinder:**
 - **Hue:** It is direction of all color on a color wheel, independent of brightness.
 - **Saturation:** It tells how intense a color is (low - grey/neutral and high - vivid.)
 - **Value:** Overall brightness (0 - black and 1 - max bright)

7 Applications in Healthcare

Digital Image Processing is widely used in healthcare.

Some important applications are:

- Tumor detection in MRI scans
- Analysis of X-ray images and CT scan
- Self driving cars
- Photo editing